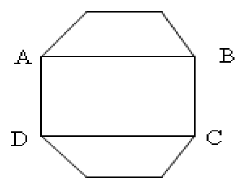
RMO(ORISSA) – 2005

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer as many questions as you can.

1.
   (a) If product of 1000 positive integers is 1000, then find the maximum sum of that 1000 positive integers.
   (b) In the figure what part of Area of regular octagon is the area of rectangle ABCD.



2.
   (a) 1, 2, 3, ……, 9 are arranged in a circle. If we take 3 continuous numbers in cyclic order we will get 9 three digit numbers. Find the sum of all the 9 three digit numbers.
   (b) X is 6 digit number. By taking left three digit of X and kept right side and make it Y (Example X = 385467, Y = 467385). Prove that when X anf Y divided by 27 they give the same remainder.

3.
   (a) Find all function f such that $f(x)f(y)-f(xy)=x+y$.
   (b) $n\geq 1$ is an integer. Prove that $(1+x)^n \geq 1+\frac{n^2}{4}$, where x is a real number and $x\geq 1$.

4. Find all real numbers x, y and z such that $xy=z-x-y,\quad xz=y-x-z,\quad yz=x-y-z$.
5. Perimeter of a triangle ABC is P and area is A and radius of its circum circle is R, then find the maximum value of $\frac{P.A}{R^3}$.
6. For any positive integer n, $f(n)=\frac{4n+\sqrt{4n^2-1}}{\sqrt{2n+1}+\sqrt{2n-1}}$, find value of $f(1)+f(2)+......+f(40)$.
7. If $S_n=(3+\sqrt{5})^n+(3-\sqrt{5})^n$ for $n=0,1,2,3......$ then prove that S_n is an integer. Also prove that integer just greater than $(3+\sqrt{5})^n$ is divisible by 2^n.
8. A Quadrilateral ABCD is circumscribed by a circle of radius 1 and $AB.BC.CD.DA\geq 4$. Prove that ABCD is a square.
9. M is an interior point of triangle ABC. Perimeter of triangle ABC is P. Prove that $\frac{P}{2}<AM+MB+MC<P$.
10. Find the sequence of real numbers x_0,x_1x_2........such that $0<x_0\leq 1,\quad 0<x_{n+1}\geq 2-\frac{1}{x_n}$.

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